

Rabiat Sadiq

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EDUCATION

Carnegie Mellon University

M.S. in Computer Science, Human Computer Interaction, concentration in AI/ML and XR.

Pittsburgh, PA

August 2026

The University of Texas at San Antonio

B.S. in Computer Engineering, Minor in Computer Science.

San Antonio, TX

December 2024

EXPERIENCE

Technical Product Manager, Design Technologist and UX Researcher

January 2026 – August 2026

Amazon Music, Carnegie Mellon MHCI Capstone

- Led mixed-methods research on music discovery and fan belonging: 74-response survey (61 variables), 8 semi-structured interviews spanning casual listeners to established artists, 10+ field intercepts, and record-store ethnography mapping tactile browsing behaviors and physical playback rituals.
- Synthesized 600+ affinity notes into a touchpoint/tension model and customer-lifecycle segmentation; ran 3 rounds of HMW reframing to narrow from feature ideation to system-level design principles.
- De-risked 5 design directions against a comprehension-by-desirability matrix; ran a Wizard-of-Oz study (5 moderated sessions) showing listeners will trade personal context for relevance when the system explains its reasoning and offers exit controls.
- Built and shipped the team's cross-platform prototype in TypeScript on React Native/Expo: 31 routes and 46 components from a single codebase targeting iOS, Android, and web; owned release management and Git workflow.
- Reverse-engineered Amazon Music's information architecture, layout, motion design, and native typography so the experimental adaptive-UI layer was the only isolated variable under test.

Graduate XR Researcher

September 2025 – August 2026

Carnegie Mellon University, Augmented Perception Lab (XR) / HCII

Pittsburgh, PA

- Build multimodal XR prototypes in Unity, investigating how visual, spatial audio, and ambient cues affect pain perception in controlled VR experiments.
- Design adaptive interfaces and log time-series user study data to support ML-driven feedback systems.
- Collaborate with interdisciplinary HCI faculty to advance human-centered XR and AI integration.

Applied Machine Learning Intern, Applied AI Team

June 2025 – August 2025

PlayStation, Sony Interactive Entertainment

Aliso Viejo, CA

- Built an end-to-end ML gameplay video context pipeline to extract on-screen scores and align outputs to timestamps for validation and downstream streaming analytics.
- Implemented parallel cloud ingestion from AWS S3 using multithreaded downloads and Ray task parallelism, reducing large video download time by 60%+.
- Automated preprocessing by splitting long-form streams into clips with FFmpeg, processing frames with OpenCV/OCR, and running scalable jobs in Databricks with PySpark.
- Tracked runs and artifacts with MLflow; improved data quality through iterative validation and performance-focused debugging.

AI/ML Program Participant

June 2024 – August 2024

Apple, AI/ML Program

Los Angeles, CA

- Built and deployed models across image classification, clustering, NLP, and responsible AI in an intensive 8-week applied ML program using Python, PyTorch, TensorFlow, CoreML, and scikit-learn.
- Engineered ML solutions using CNNs, transfer learning, SVMs, and decision trees; evaluated models with accuracy, precision, recall, F1 score, and ROC-AUC.
- Applied Swift and Xcode for on-device ML inference; leveraged linear algebra, statistical modeling, and feature engineering throughout hands-on projects.

AI/VR Undergraduate Research Assistant

August 2023 – August 2024

UTSA

San Antonio, TX

- Researched multimodal AI moderation for social VR safety, building Unity/C# data collection tooling and developing supervised and LLM-based behavior classification to reduce harassment and unsafe interactions.

Software Engineer Participant, Software Product Sprint Program

Summer 2022 – Summer 2022

Google

Remote

- Built Talky Talky, an audio-responsive web application for non-verbal children, using Java, JavaScript, HTML, CSS, and Google Text-to-Speech APIs with Google Cloud Platform services.

PROJECTS

Xbox x CMU: Measuring Social in Gaming

January 2026 – August 2026

AI Research Engineer

- Partner with Xbox on an AI research project to measure and understand social interactions in gaming environments using AI-driven approaches.
- Develop methods to quantify social dynamics and player behavior to inform intelligent, human-centered gaming systems.

Magic Mitts: Affordable, Immersive VR Gloves

January 2024 – December 2024

Lead Project Manager and Computer Engineer

- Invented a smart, programmable VR glove using a novel electromagnetic braking system and haptic actuators to deliver precise tactile feedback, simulating touch and grasp across standard VR systems.
- Developed algorithms for sensor processing, haptic control, and feedback synchronization; integrated flex-sensitive resistors for finger-tracking accuracy and a self-developed hardware driver for wireless, rechargeable operation.
- Placed 1st out of 89 teams at the UTSA Tech Symposium.

Sensing + ML Distress Prediction

- Built an end-to-end iOS sensing-to-insight pipeline transforming Apple HealthKit biometrics (HR, HRV, respiratory rate, steps, active energy, BMI) into real-time distress predictions.
- Trained and evaluated a logistic regression model achieving 82% test accuracy using pseudo-data augmentation; integrated on-device CoreML inference in Swift for live in-app feedback.

Applied STEM (Startup)

Co-founder, AI and Full-Stack

- Built a context-aware circuit interview mentor (React/TypeScript + FastAPI) that grounds LLM feedback in live circuit canvas state and simulation outputs to deliver adaptive questions and progressive hints.

SKILLS

ML and Computer Vision: Python, PyTorch, TensorFlow, scikit-learn, OpenCV, CoreML, CNNs, transfer learning, RNNs, embeddings, clustering, PCA, SVM, Computer Vision, EDA, model evaluation, feature engineering, cross-validation, Matplotlib/Seaborn, Image Processing, Object Detection

XR Engineering: Unity (C#), Meta XR SDK, VR prototyping, interaction systems, debugging

Video and Data Pipelines: FFmpeg, Ray, Databricks, PySpark, MLflow, SQL, Snowflake, multithreading, concurrency, Kafka

Engineering and Tooling: TypeScript, React Native, Expo, React, Swift, Xcode, Docker, Git/GitHub, CI/CD, Jupyter Notebooks, Jira, Confluence

Research Methods: Semi-structured interviews, contextual inquiry, participatory audits, ethnographic field research, survey design and analysis, affinity diagramming, Wizard-of-Oz prototyping, usability testing

Product and Design: Roadmapping and scoping, prioritization frameworks, stakeholder communication, release management, Figma, design systems, information architecture, rapid prototyping

Hardware: Microcontrollers, sensors and signal processing, haptics, electromagnetics, circuit design and analysis, embedded systems, Verilog, MATLAB

ACTIVITIES & LEADERSHIP

Professional Organizations

- IEEE, SWE, NACME, NSBE, RAAS (Robotics Association at UTSA)

AWARDS & HONORS

- Apple Scholar Program (January 2022)
- Horatio Alger Association Texas Scholarship (February 2021 and June 2023)
- Michael and Susan Dell Scholarship (March 2021)
- Marge and Bill Kleese ECE Scholarship (2022-2023)